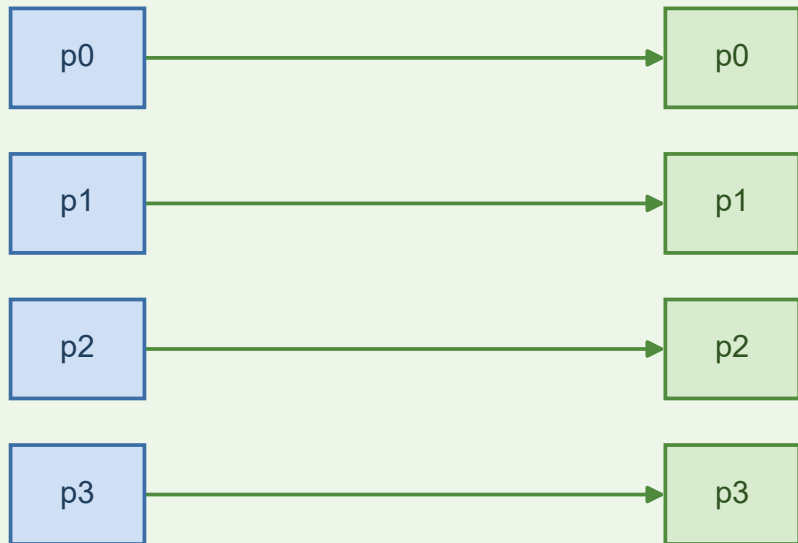


Narrow dependency

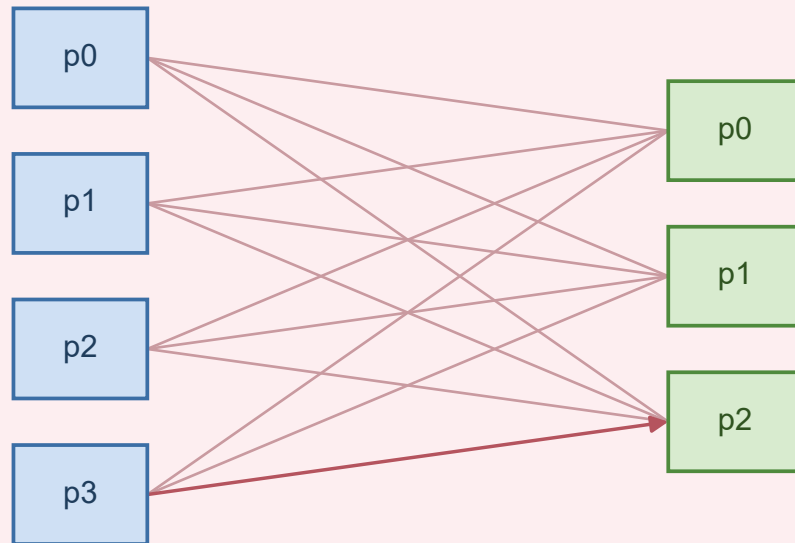
filter • project • map



each output partition reads one input • no data moves • stays in the stage

Wide dependency

group by • join • repartition



every output reads every input • the shuffle • ends the stage